STACKS, DIFFERENTIAL GRADED SHEAVES: LEMMA-SMALL-ACYCLICS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3851-3860 of sources/stacks-sdga/pinned-original-source.tex; proof span is lines 3862-3949, the complete author proof (86 body lines).

Editorial note. Navigation stubs: 0 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 4 of queue_sdga.json (pb=3601, stmt 3851-3860, proof 3862-3949).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let $(\mathcal{A}, d)$ be a sheaf of differential graded algebras on $(\mathcal{C}, \mathcal{O})$. There exists a set S and for each s an acyclic differential graded $\mathcal{A}$-module $\mathcal{M}_s$ such that for every nonzero acyclic differential graded $\mathcal{A}$-module $\mathcal{M}$ there is an $s \in S$ and an injective map $\mathcal{M}_s \to \mathcal{M}$ in Mod$(\mathcal{A}, d)$.*

Proof. Before we start recall that our conventions guarantee the site $\mathcal{C}$ has a set of objects and morphisms and a set $\mathrm{Cov}(\mathcal{C})$ of coverings. If $\mathcal{F}$ is a differential graded $\mathcal{A}$-module, let us define $|\mathcal{F}|$ to be the sum of the cardinality of

$$\coprod\nolimits_{(U,n)} \mathcal{F}^n(U)$$

as U ranges over the objects of $\mathcal{C}$ and $n \in \mathbf{Z}$. Choose an infinite cardinal κ bigger than the cardinals $|\mathrm{Ob}(\mathcal{C})|$, $|\mathrm{Arrows}(\mathcal{C})|$, $|\mathrm{Cov}(\mathcal{C})|$, $\sup |I|$ for $\{U_i \to U\}_{i \in I} \in \mathrm{Cov}(\mathcal{C})$, and $|\mathcal{A}|$.

Let $\mathcal{F} \subset \mathcal{M}$ be an inclusion of differential graded $\mathcal{A}$-modules. Suppose given a set K and for each $k \in K$ a triple (U_k, n_k, x_k) consisting of an object U_k of $\mathcal{C}$, integer n_k, and a section $x_k \in \mathcal{M}^{n_k}(U_k)$. Then we can consider the smallest differential graded $\mathcal{A}$-submodule $\mathcal{F}' \subset \mathcal{M}$ containing $\mathcal{F}$ and the sections x_k for $k \in K$. We can describe

$$(\mathcal{F}')^n(U) \subset \mathcal{M}^n(U)$$

as the set of elements $x \in \mathcal{M}^n(U)$ such that there exists $\{f_i : U_i \to U\}_{i \in I} \in \mathrm{Cov}(\mathcal{C})$ such that for each $i \in I$ there is a finite set T_i and morphisms $g_t : U_i \to U_{k_t}$

$$f_i^* x = y_i + \sum\nolimits_{t \in T_i} a_{it} g_t^* x_{k_t} + b_{it} g_t^* \mathrm{d}(x_{k_t})$$

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for some section $y_i \in \mathcal{F}^n(U)$ and sections $a_{it} \in \mathcal{A}^{n-n_{k_t}}(U_i)$ and $b_{it} \in \mathcal{A}^{n-n_{k_t}-1}(U_i)$. (Details omitted; hints: these sections are certainly in $\mathcal{F}'$ and you show conversely that this rule defines a differential graded $\mathcal{A}$-submodule.) It follows from this description that $|\mathcal{F}'| \leq \max(|\mathcal{F}|, |K|, \kappa)$.

Let $\mathcal{M}$ be a nonzero acyclic differential graded $\mathcal{A}$-module. Then we can find an integer n and a nonzero section x of $\mathcal{M}^n$ over some object U of $\mathcal{C}$. Let

$$\mathcal{F}_0 \subset \mathcal{M}$$

be the smallest differential graded $\mathcal{A}$-submodule containing x. By the previous paragraph we have $|\mathcal{F}_0| \leq \kappa$. By induction, given $\mathcal{F}_0, \ldots, \mathcal{F}_n$ define $\mathcal{F}_{n+1}$ as follows. Consider the set

$$L = \{(U, n, x)\}\{U_i \to U\}_{i \in I}, (x_i)_{i \in I})\}$$

of triples where U is an object of $\mathcal{C}$, $n \in \mathbf{Z}$, and $x \in \mathcal{F}_n(U)$ with $\mathrm{d}(x) = 0$. Since $\mathcal{M}$ is acyclic for each triple $l = (U_l, n_l, x_l) \in L$ we can choose $\{(U_{l,i} \to U_l\}_{i \in I_l} \in \mathrm{Cov}(\mathcal{C})$ and $x_{l,i} \in \mathcal{M}^{n_l - 1}(U_{l,i})$ such that $\mathrm{d}(x_{l,i}) = x|_{U_{l,i}}$. Then we set

$$K = \{(U_{l,i}, n_l - 1, x_{l,i}) \mid l \in L, i \in I_l\}$$

and we let $\mathcal{F}_{n+1}$ be the smallest differential graded $\mathcal{A}$-submodule of $\mathcal{M}$ containing $\mathcal{F}_n$ and the sections $x_{l,i}$. Since $|K| \leq \max(\kappa, |\mathcal{F}_n|)$ we conclude that $|\mathcal{F}_{n+1}| \leq \kappa$ by induction.

By construction the inclusion $\mathcal{F}_n \to \mathcal{F}_{n+1}$ induces the zero map on cohomology sheaves. Hence we see that $\mathcal{F} = \bigcup \mathcal{F}_n$ is a nonzero acyclic submodule with $|\mathcal{F}| \leq \kappa$. Since there is only a set of isomorphism classes of differential graded $\mathcal{A}$-modules $\mathcal{F}$ with $|\mathcal{F}|$ bounded, we conclude. $\square$